208 Projectile Motion

2.8.1 Resolving projectile motion

Neglecting air resistance, a projectile has only downward gravitational acceleration. Its horizontal velocity is constant while its vertical motion has uniform acceleration. Both components share the same flight time but can be calculated separately. At the highest point only vertical velocity is zero; horizontal velocity and downward acceleration generally remain non-zero.

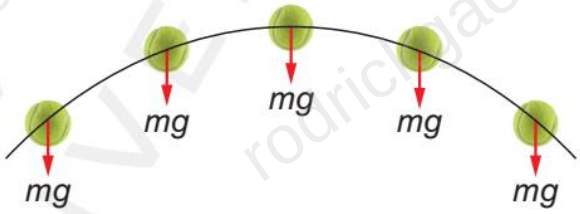


Fig 8.1c *A projectile is acted on only by the gravitational force mg.*

Gravity acts downward at every point, including the highest point. With negligible air resistance, horizontal velocity stays constant.

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